

Multi-objective optimization approach for green design of methanol plant based on CO₂-efficiency indicator



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ABSTRACT

The aim of the present work is to propose an eco-design method for sustainable development of methanol production by implementing a multi-objective optimization model based on CO₂-efficiency. Two different approaches for the methanol production, i.e. a conventional reference methanol case (RMC) and proposed green integrated methanol case (GIMC) were compared from the view point of eco-design. Using life cycle assessment and process simulation, the environmental features as well as operational decision variables of the RMC and GIMC were assessed. Based on the inventory analysis of LCA, it was found that carbon dioxide is the major emitted pollutant from methanol production. Thus the multi-objective model in the proposed GIMC was formulated as *maximizing methanol production* (which refers to a classical objective in process design) and *minimizing CO₂ emission* (which refers to an environmental objective in process design) and the set of Pareto frontier was obtained. In order to achieve single optimal solution, the multi-objective model is transformed to a single objective optimization problem by weighted product method and the single optimum point is found by using genetic algorithm. Comparing obtained results showed that the optimal condition of the RMC which is based on a classical objective is a local optima from the view point of environmental benign design, whereas by involving the environmental objective in the design stage, a lower CO₂ emission with respect to the RMC could be achieved at the expense of a small decrease in the methanol production.

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1. Introduction

The general concept of eco-design is an essential way to reduce undesired environmental impacts of processes, systems and products. Traditional process design often pays attention to economic criteria such as fixed capital cost, operating cost and rate of return on investment (RRI). In the past, the environmental impacts of process design had a lower priority and generally were incorporated into traditional design as an end-of-pipe treatment (Li et al., 2009). During the last decade, because of the growing awareness about the importance of environmental concerns, environmental management strategies have been shifted from reactive to proactive measures for sustainable development (Lim and Park, 2009). In this regard, concepts like green design, cleaner production, environmental benign design and eco-design have been used to improve the environmental and economic performances of products and processes. These concepts have

commonly been implemented by two approaches: treating the environmental considerations as constraints or legislations that aim to mitigate environmental impacts, or dealing with the environmental problems as a separate objective in the process synthesis stage and process selection. In the latter approach environmental requirements are considered as a separate objective together with other classical objectives such as maximizing profit or maximizing production of the desired product to form a multi-objective optimization (MOO) problem. For example Li et al. have used a multi-objective mixed integer non-linear mathematical model by considering environmental and economic factors (Li et al., 2009). Tian et al. have utilized a multi-objective optimization model to consider both the energy consumption as well as the environmental impact (green degree) of a proposed new ionic liquid process to produce 1,3-butadiene (Tian et al., 2011). In their model a nonlinear mathematical model with six decision variables was established to express the MOO problem, which involves maximizing the green degree of the process, the purity and the recovery of 1,3-butadiene, and minimizing the energy consumption of the process. Sayyaadi et al. applied a multi-objective optimization approach in which exergetic, exergoeconomic and

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Abbreviations / nomenclature

ATAP	atmospheric acidification potential (tonne/yr SO ₂ equivalent)	MOO	multi-objective optimization
ATR	autothermal reactor	n	number of independent variables
CASU	cryogenic air separation unit	NGF	natural gas feed (tonne/h)
CFC-11	trichlorofluoromethane	NGFF	portion of natural gas feed used as fuel in GIMC (tonne/hr)
$F(\vec{x})$	vector of objective functions	NGFS	portion of natural gas feed allocated for syngas production in GIMC (tonne/hr)
$f_{eq}(\vec{x})$	composite function	OLE	object Linking and Embedding
$f_i(\vec{x})$	i th objective function	p	number of equality constraints
F^*	vector of Pareto optimal points	POFP	photochemical ozone or smog formation potential (tonne/yr ethylene equivalent)
$(\dot{F}_{CO_2}^1)$	injected carbon dioxide mass flow to the steam reformer reactor (tonne/h)	RMC	reference methanol case
$(\dot{F}_{CO_2}^2)$	Injected carbon dioxide mass flow to methanol synthesis reactor (tonne/h)	RRI	rate of return on investment
$(\dot{F}_{steam})$	additional injected steam mass flow (tonne/hr)	S	feasible criterion space
GA	genetic algorithm	SDT	social disaster technique
GIMC	green integrated methanol case	SODP	stratospheric ozone depletion potential (tonne/yr CFC-11 equivalent)
GTPP	gas turbine power plant	SOO	single objective optimization
GWP	global warming potential (tonne/yr CO ₂ equivalent)	SRR	syngas recycle ratio (%)
$\vec{g}(\vec{x})$	vector of inequality constraints	SRT	steam reformer temperature (°C)
$h(\vec{x})$	vector of equality constraints	S_R	stoichiometric ratio of synthesis gas (dimensionless)
HHP	human health or carcinogenic effect (tonne/yr benzene equivalent)	VOCs	volatile organic compounds
k	number of objectives	w	vector of weights (or powers) (dimensionless)
LCA	life cycle assessment	w_i	weight/power factor for objective function f_i (dimensionless)
m	number of inequality constraints	WPM	weighted product method
M	stoichiometric ratio of synthesis gas (dimensionless)	WSM	weighted sum method
MeOH	methanol	$\vec{x}$	vector of decision variables
MOM	multi-objective model	X	feasible decision space

environmental objectives are considered. In their work, exergetic efficiency as an exergetic objective was maximized while the unit cost of the system product and the cost of environmental impact respectively as exergoeconomic and environmental objectives were minimized (Sayyaadi et al., 2011).

This approach is obviously more powerful; however, it is difficult to find a suitable mathematical solution because of the highly complex nature of the process and the difficulties of developing a mathematical model (Li et al., 2009). Since a MOO problem requires simultaneous satisfaction of different and often conflicting objectives, there is no single global solution and it is necessary to determine a set of points that represent the optimal trade-off between different objectives, i.e. Pareto frontier. All points on the Pareto frontier are mathematically equal and there is no preference among different solutions on a Pareto set. Although there are many choices on Pareto frontier, from a practical point of view, a user needs one solution, no matter if the optimization problem is a single or multi-objective (Deb, 2005). There are two approaches to find a single solution in multi-objective optimization problems, i.e. methods with a prior preferences (or preference-based procedures) and methods with a posterior preferences (or ideal procedures) (Marler and Arora, 2004). In prior preferences methods, one can estimate the relative importance of different objective functions by using higher level information. Based on this estimation, the multi-objective optimization problem could be transferred into a single objective optimization problem (SOO) and the new SOO problem is solved by a classical or an evolutionary method. For example, the most common method in prior preferences approach is weighted sum method (WSM) in which transferring multi-objective optimization problem into a single objective optimization problem is accomplished by pre-multiplying each objective with a user-

supplied weight factor (Deb, 2005). In this method, the assignment of the weight factors to different objectives is under influence of decision makers' opinion and based on their preferences, different solutions could be achieved for the same problem. As could be observed, in prior preferences methods before viewing the single solution point, one must quantify the relative importance of different objectives (Marler and Arora, 2004). Vice versa, in posterior preferences methods, decision makers implement their preferences directly on the Pareto frontier and choose one of the equivalent options as a best single solution point. In this method, the final solution accurately reflects the decision makers' preferences (Marler and Arora, 2004).

Another important factor in multi-objective approach to the environmental problems is how to find a suitable environmental objective which represents the whole environmental impacts of the process (Li et al., 2009). For example, in the early 1990s, waste minimization techniques were applied for optimization of environmental performance along with conventional criteria (El-Halwagi and Manousiouthakis, 1990; Ciric and Jia, 1994; Young and Cabezas, 1999). However, the main disadvantage of this approach was to focus on the emissions from the main plant alone without considering other stages in the life cycle. Thus, it is possible to reduce the emissions from the plant while the burdens increase elsewhere in the life cycle, so that the overall environmental impacts are increased (Azapagic and Clift, 1999a). However, there are several systematic methodologies such as Life Cycle Assessment (LCA) in which the environmental impacts of the main plant as well as those of other stages in the life cycle is considered in order to identify and quantify potential environmental burdens and impacts of a process in a comprehensive approach (Finnveden et al., 2009; ISO, 2006a; ISO, 2006b; Lou et al., 2004). Although the application

of LCA provides valuable information about the environmental problems, it does not include a systematic method of introducing environmental friendly alternative processes for environmental improvements. This drawback can be overcome by using the LCA results within the design stage in the multi-objective optimization approach (Azapagic and Clift, 1999b; Finnveden et al., 2009).

The aim of the present work is to propose an eco-design method for sustainable development of the methanol production by implementing a multi-objective optimization model based on CO₂-efficiency Indicator. As a starting point, in Section 2, some general concepts such as life cycle assessment, methanol plant technologies and multi-objective optimization will be introduced. In Section 3, an integrated methodology for development of a multi-objective optimization model is described where creating a multi-objective model, the Pareto optimal frontier and transferring multi-objective optimization model to a single objective optimization model are discussed in detail. The pre-optimization and sensitivity analysis as well as the LCA and inventory analysis of the RMC and the GIMC is performed in Section 4. This is followed by solving the multi-objective optimization problem with weighted product method and eco-efficiency concept. Assessing premature convergence as well as trapping in a local optimal point is also assessed. Finally the conclusion is presented in section 5. It is worth mentioning that in the field of methanol production; some attempts have been made to optimize methanol production focusing on the improvement of catalyst performance or proposing different reactor types. For example Riaz et al. summarized cleaner production methods for methanol manufacturing (Riaz et al., 2013). Van-Dal et al. proposed a process for producing fuel grade methanol from captured CO₂ in which CO₂ is provided from the flue gas of a power plant while hydrogen is produced by water electrolysis (Van-Dal and Bouallou, 2013). Furthermore, in our previous work (Taghdisian et al., 2012), GIMC ability for production of syngas with $S_R = 2$ was assessed where CO₂ was used as an adjusting agent in order to optimize the synthesis gas composition. It was shown that the proposed methanol case (with CO₂ as an adjusting agent of synthesis gas composition) was capable of producing the same amount of syngas and methanol as that of the RMC. To the best of our knowledge, multi-objective optimization approach to the environmental aspects of CO₂ injection in methanol plant have not been addressed in the literature. In the present work, it is tried to maximize the methanol production as well as to minimize the most important environmental burden of methanol plant.

2. General concepts and methods

2.1. Life cycle assessment

LCA is an environmental procedure which is used to systematically evaluate the potential of environmental impacts and burdens. A life cycle assessment consists of four steps: Goal and scope definition, Inventory analysis, Impact assessment and Improvement Assessment (Azapagic and Clift, 1999a; ISO, 2006a). In the first step, the system boundaries are defined to ensure that all relevant parts of the system are considered. The boundary of the system should include all material and energy streams in the entire life cycle of the process being analyzed. In the second step of LCA, energy and mass balance are performed to quantify all the material and energy inputs, waste and emissions from the system (i.e., the environmental burdens).

In the third step, the calculated environmental burdens in the life cycle inventory step are aggregated into three main categories (i.e. Atmospheric Impacts, Aquatic impacts and Land impact) where each category consists of several impacts. The main advantage of the life cycle assessment amongst environmental impact

assessment methods lies in this step where the environmental burdens and impacts are provided in a quantitative basis. This quantitative nature of LCA introduces it as a powerful tool for improvement of the environmental performance (Azapagic and Clift, 1999a; Hojer et al., 2008).

In the last step of LCA, assessment improvement is performed to evaluate different options and scenarios to choose the best solution. In this framework, LCA is employed to assess different technological options from the view point of environmental benign design, whereas optimization tool searches for the best solution according to the predefined constraints and criteria. This approach is used in the present work where quantitative information of second and third steps of LCA will be used in multi-objective optimization formulation and the optimization engine searches for the best solution to achieve more environmentally friendly process.

2.2. Methanol plant description (RMC and GIMC)

Methanol technologies feature three process sections: Syngas preparation (or reforming unit), methanol synthesis and methanol purification blocks. The synthesis gas preparation and compression sections typically account for more than 60% of the capital investment cost of a methanol plant (Aasberg-Petersen et al., 2001). Almost all energy demand in the methanol production is consumed in the synthesis gas reforming unit. Many investigations have been focused on optimization of syngas process scheme specially for avoiding catalyst deactivation with a few studies considering optimization of methanol plant from the view point of environmental concerns (Riaz et al., 2013). Table 1 shows main synthesis gas reactions in steam reforming, CO₂-reforming and autothermal reforming.

Methanol synthesis gas is characterized by the stoichiometric ratio, often referred to as module M or S_R number (Eq. (1)), where the desirable value for methanol production is $S_R = 2$ (Rostrup-Nielsen, 2000).

$$S_R = \frac{H_2 - CO_2}{CO + CO_2} = 2.0 - 2.1 \quad (1)$$

The module $M = 2$ is an operational objective corresponding to a value close to stoichiometric condition for methanol production where the maximum amount of methanol could be achieved. If the module M is larger than its optimum value, unreacted hydrogen is recycled back to the reformer as a purge gas fuel whereas below stoichiometric ratio (i.e. $M < 2$) formation of by-products such as higher alcohols increases (Riaz et al., 2013). Therefore, to achieve the maximum methanol production (or profit), as a traditional objective in methanol synthesis, the module M is set at 2 or slightly above it. The choice of syngas technology to obtain this goal is dictated by the scale of operation (Rostrup-Nielsen, 2000). Since the steam reforming produces surplus hydrogen ($M > 2$), syngas

Table 1
Syngas reactions (Elnashaie and Elshishini, 1993; Rostrup-Nielsen, 2000).

	ΔH_{298}^0 (kJ/mol)
<i>Steam reforming</i>	
$CH_4 + H_2O \leftrightarrow CO + 3H_2$	206
$CO + H_2O \leftrightarrow CO_2 + H_2$	−41
$C_nH_m + nH_2O \leftrightarrow nCO + (n + m/2)H_2$	−
<i>CO₂-reforming</i>	
$CH_4 + CO_2 \rightarrow CO + 2H_2$	247
<i>Autothermal reforming</i>	
$CH_4 + 1.5O_2 \rightarrow CO + 2H_2O$	−520
$CH_4 + H_2O \leftrightarrow CO + 3H_2$	206
$CO + H_2O \leftrightarrow CO_2 + H_2$	−41

with $M = 2$ is produced either by two-step reforming or CO_2 -reforming. In two-step (or combined) reforming, a tubular steam reformer followed by an autothermal reformer is employed whereas in CO_2 -reforming, CO_2 injection into the reforming unit is utilized to adjust the stoichiometric ratio. It should be noted that CO_2 -reforming feasibility strongly depends on the price of CO_2 source (Rostrup-Nielsen, 2000). This approach is feasible for natural gas feedstock which contains some CO_2 or in case of an available cheap source of CO_2 (Holm-Larsen, 2001; Rostrup-Nielsen, 2000).

2.2.1. Reference methanol case (RMC)

A simplified schematic flow diagram of the reference methanol plant with two-step reforming is shown in Fig. 1. After desulfurization of natural gas feed, steam is added to the desulfurized natural gas and the mixture is divided into two streams. The first stream (S1) is directed toward the pre-reformer and steam reformer while the second stream (S2) is added to the first one before entering autothermal reformer. In the adiabatic pre-reformer, all higher hydrocarbons (i.e. C_2^+) are converted into a mixture of methane, hydrogen and carbon oxides (reaction 3 in Table 1) (Aasberg-Petersen et al., 2001). The pre-reformer outlet then enters the steam reformer which is a fired tubular reactor. Methane conversion occurs in the steam reformer via an endothermic catalytic reaction and provides a mixture of H_2 , CO , CO_2 , H_2O and non-reacted CH_4 . Typically, H_2/CO ratios between 3 and 5 would be achieved depending on the feedstock, CO_2 recycle and operating conditions. The reformed gas is then mixed with the rest of the feed gas (i.e. S2 stream) and reformed to final synthesis gas in an autothermal reactor (ATR). In the ATR, the heat of reaction required for steam reforming of methane is supplied directly by partial oxidation of hydrocarbons using pure oxygen in the upper section of the reactor. The catalytic bed in the lower section of ATR ensures the production of a mixture with the required gas composition close to the equilibrium. The huge amount of pure oxygen needed in autothermal reformer is produced in a separate cryogenic air separation unit (CASU). Cryogenic air separation is a highly energy intensive process so that up to 40% of the costs of a syngas plant in two-step reforming is related to the oxygen plant

(Rostrup-Nielsen, 2002, 2000). In the RMC, the electricity consumption for oxygen production by cryogenic distillation of air is supplied by a separate gas turbine power plant.

Crude methanol which is a mixture of methanol, a small amount of water, dissolved gases, and traces of by-products is generated from synthesis gas in a catalytic reactor. Vanden Bussche and Froment (1996) proposed a steady-state kinetic model for methanol synthesis (Table 2). Although the methanol synthesis catalyst is highly selective, methanol production is strongly influenced by thermodynamics (Lange, 2001). Thus, recycling of the unconverted gases should be provided if a high conversion is required. In order to produce pure methanol (i.e. Grade AA), the crude methanol is purified in a three column distillation train in which the low boiling components are removed in the pre-run column while the higher boiling components are separated within the next two columns.

In order to obtain mass and energy information as well as pollutant emission inventory and environmental impacts of the RMC, the RMC was modeled in Aspen HYSYS. Using several case studies and sensitivity analysis, operating conditions and kinetic parameters of the RMC were estimated and verified with respect to the operational data of one of the active methanol production plant in Iran. The simulated plant is able to accurately describe the actual behavior of the currently active methanol plant. By defining several spreadsheets in HYSYS which are dynamically linked together, mass and energy information, inventory analysis and environmental impacts are calculated and the results are exported to Excel using Object Linking and Embedding (OLE). Using practical data and linking Excel Visual Basic for Applications and HYSYS simulator, operating conditions as well as environmental burdens of the RMC were calculated.

2.2.2. Green integrated methanol case (GIMC)

Another approach to achieve synthesis gas with module $M = 2$ is combined steam and CO_2 -reforming where its feasibility strongly depends on the price of the CO_2 source (Aasberg-Petersen et al., 2001). In the proposed green integrated methanol case, the source of CO_2 is the methanol plant itself where injected CO_2 is supplied from reformer flue gas. As shown in Fig. 2, in order to capture the required CO_2 from the reformer flue gas, the flue gas is

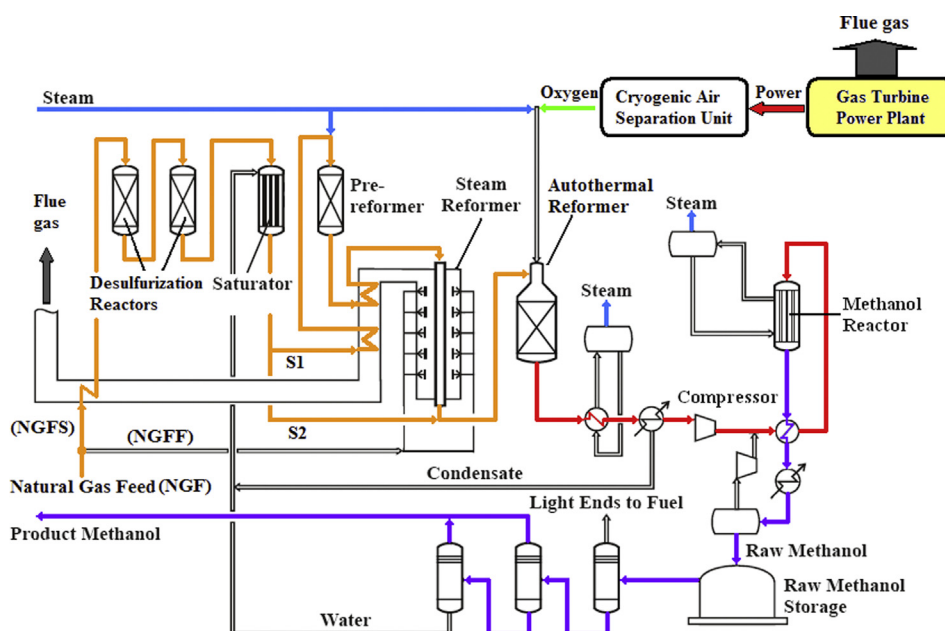


Fig. 1. Schematic flow diagram of reference methanol case in the present study (Taghdisian et al., 2012).

Table 2
Methanol formation reactions (Vanden Bussche and Froment (1996)).

	ΔH_{298}^0 (kJ/mol)
$\text{CO}_2 + 3\text{H}_2 \leftrightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$	−49.3
$\text{CO}_2 + \text{H}_2 \leftrightarrow \text{CO} + \text{H}_2\text{O}$	41.1

treated in an absorption plant and the purified CO_2 is then pressurized and used in the methanol process. There are several commercially available process technologies which can be used for CO_2 capture from flue gases. Comparative assessment studies have shown that absorption processes based on chemical solvents are not only the preferred option for post-combustion CO_2 capture, but also feasible for large scale CO_2 recovery plants (Meerman et al., 2012; Metz et al., 2005; Olajire, 2010; Yagi et al., 2004; Zhao et al., 2012). Similar to the RMC, the GIMC is modeled based on our previously verified model of the RMC to obtain its mass and energy information as well as its life cycle inventory and environmental impacts of the emissions. The process and environmental data are also exported to Excel using OLE.

2.3. Multi-objective optimization (MOO)

Multi-objective, multi-criterion or vector optimization is a process in which a number of objective functions - commonly conflicting - are simultaneously optimized. This problem can be mathematically formulated as below:

$$\begin{aligned} &\text{minimize } F(\vec{x}) = [f_1(\vec{x}), f_2(\vec{x}), \dots, f_k(\vec{x})]^T \\ &\text{Subject to :} \\ &\vec{g}(\vec{x}) \leq 0 \\ &\vec{h}(\vec{x}) = 0 \end{aligned} \quad (2)$$

$\vec{x} \in \mathbb{R}^n$, $\vec{f}(\vec{x}) \in \mathbb{R}^k$, $\vec{g}(\vec{x}) \in \mathbb{R}^m$ and $\vec{h}(\vec{x}) \in \mathbb{R}^p$

where the integer $k \geq 2$ is the number of objectives, m is the number of inequality constraints and p is the number of equality constraints. $\vec{x} \in \mathbb{R}^n$ is a vector of design or decision variables where n is the number of independent variables and $F(\vec{x}) \in \mathbb{R}^k$ is a vector of objective functions in which $f_i(\vec{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^1$. The feasible design space X (or feasible decision space) and feasible criterion space S are defined as: $X = \{ \vec{x} | g_m(\vec{x}) \leq 0, m = 1, 2, \dots, m \text{ and } h_p(\vec{x}) = 0, p = 1, 2, \dots, p \}$ and $S = \{ F(\vec{x}) | \vec{x} \in X \}$, respectively. In contrast to single objective

optimization problems in which a single solution could be achieved, typically in multi-objective optimization problems, there is no single global solution. Therefore, in such cases mathematical theory indicates a set of compromised trade-off solutions, i.e., Pareto set or Pareto frontier defined as follows: vector F is a Pareto optimal point if there does not exist another F in the feasible criterion space such that $f_i(\vec{x}) \leq f_i^*(\vec{x})$ for all $i \in \{1, 2, \dots, k\}$ and $f_i(\vec{x}) < f_i^*(\vec{x})$ for at least one $i \in \{1, 2, \dots, k\}$. Thus, for a given Pareto solution A, it is impossible to find another solution B so that it performs better than A for all of the objectives (Gebresslassie et al., 2009). Indeed all points on the Pareto frontier are mathematically equal introducing many options to decision makers who can find a single solution point by using prior or posterior preferences methods (Madetoja et al., 2008; Marler and Arora, 2004). Fig. 3 shows a general concept of feasible design space and Pareto frontier for a MOO problem with two or three objectives. Fig. 4 shows Pareto set for four possible combinations of the two types of objectives.

Classically, a multi-objective optimization model may be scalarized or transferred into a single objective optimization problem. Two simple methods in this approach are weighted sum method (WSM) and weighted product method (WPM). In the former method the composite function is generated by pre-multiplying each objective with a user-supplied weight (Eq. (3)) while in the latter method the composite function is generated by multiplication of objectives with a user-supplied power (Eq. (4)) (Deb, 2005). The mentioned weights and powers usually account for the preference paradigm in multi-criterion optimization.

$$\text{minimize } F_{\text{eq}}(\vec{x}) = \sum_{i=1}^k w_i f_i(\vec{x}) \quad (3)$$

$$\text{minimize } F_{\text{eq}}(\vec{x}) = \prod_{i=1}^k [f_i(\vec{x})]^{w_i} \quad (4)$$

In Eqs. (3) and (4) w is a vector of weights (or powers), typically set by decision makers. Both methods have their advantages and limitations. For example, although the WSM is the simplest and probably the most widely used approach, selecting appropriate values for weight factors is still another milestone for decision makers. In the case of WPM, increasing nonlinearities potential in the composite function prevents widespread use of this method. This is the case, where classical solvers are used, otherwise

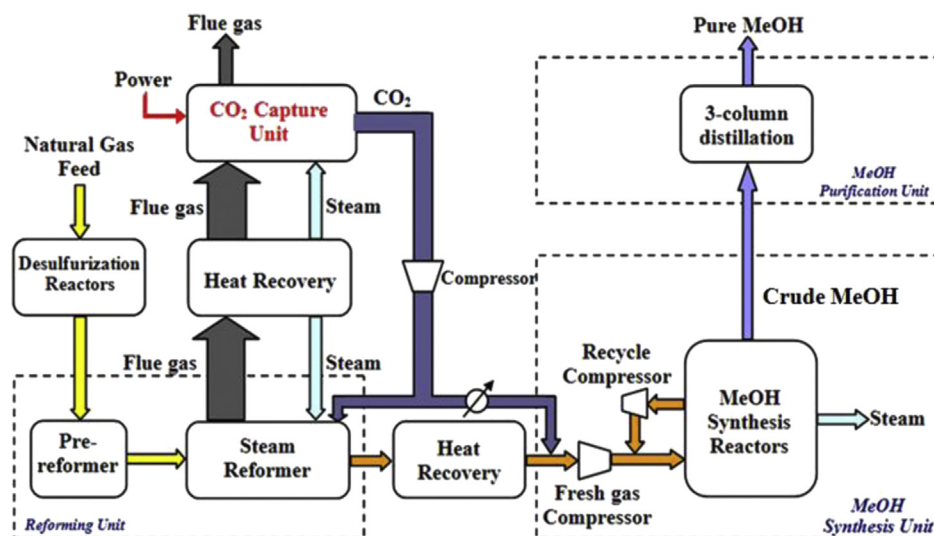


Fig. 2. Schematic flow diagram of the green integrated methanol case.

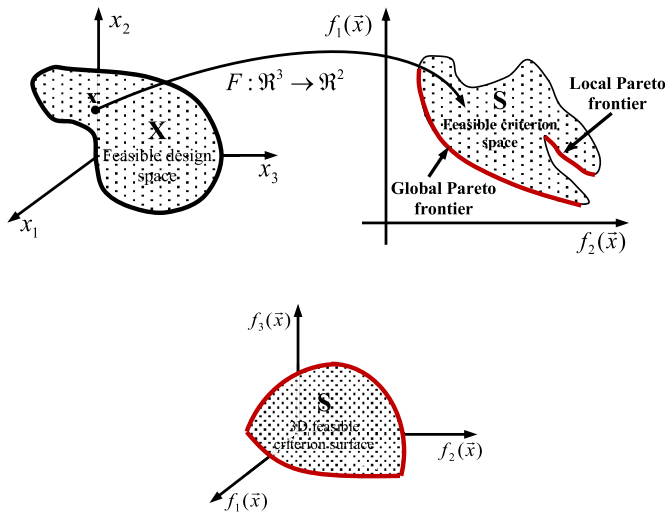


Fig. 3. Top: feasible decision space and Pareto frontier for MOO problem with two objectives. Bottom: a 3D Pareto surface for a three objective optimization problem.

evolutionary optimizers can handle the nonlinearities very dexterously.

The WPM is often used for minimization (or maximization) problems, where in the minimization strategy, objectives to be maximized can be included in the composite function by using $-w_i$ instead of w_i . On the other hand, in the maximization problems, the objective functions need to be minimized can be introduced into the composite function by using $-w_i$. Eq. (4) can also be applied without any concern for positive objective functions. However, negative (or zero) values should be avoided as exponentiating by a rational exponent may lead to a complex number. If an objective function takes only negative values $-f_i$ can be used in Eq. (4) (Marler and Arora, 2004).

In the present study, because the multi-objective formulation is a mixed optimization problem (min–max problem), WPM is used to transform the multi-objective (vector) optimization problem into a single (scalar) objective optimization problem. As will be shown, based on the eco-efficiency concept, the composite function

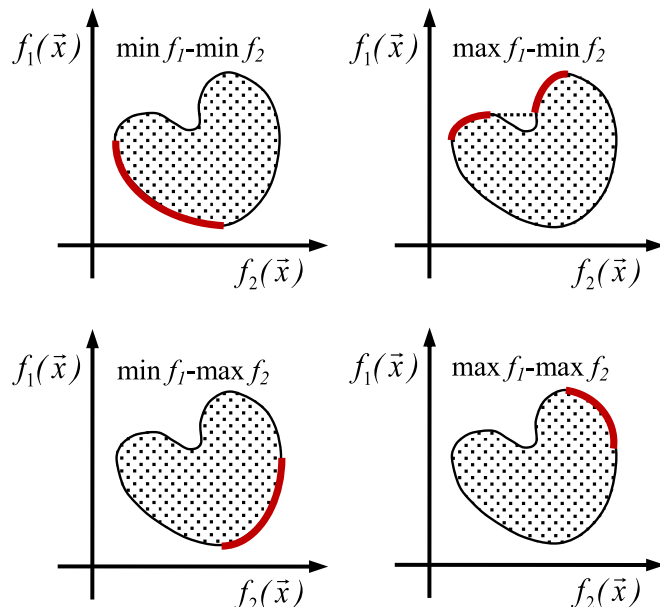


Fig. 4. Pareto set for four combinations of two types of objectives.

generated by Eq. (4) leads to an important environmental indicator, called CO₂-efficiency. Several environmental indicators such as Return on Environment (Hunkeler and Biswas, 2000), Green Productivity index (Hur et al., 2004), Sustainable development indicator (Palme et al., 2005) and CO₂-efficiency (Tahara et al., 2005) have been proposed by researchers to estimate environmental as well as economic performances. CO₂-Efficiency is based on the concept of generating more goods and services while using fewer resources and creating less waste and pollutants (Verfaillie and Bidwell, 2000). Environmental indicators such as CO₂-efficiency can help designers, manufacturers, and managers to screen out more environmental benign designs and processes. Such indicators can also be used for providing qualitative and quantitative information on environmental performance compare alternative designs (Lim and Park, 2009). By introducing CO₂-efficiency as a standard indicator, it is possible to compare environmental performance of different industries for their contribution in the global warming effect (Tahara et al., 2005).

3. Integrated methodology for MOO

In the present study, the integrated methodology to develop a MOO model contains three main steps: creating a multi-objective model, generating Pareto optimal frontier and transferring the MOO model to a SOO model to achieve the single solution point. Fig. 5 shows the schematic procedure of the integrated methodology used in this study. In the first step, by using the environmental model (i.e., LCA) and process model (i.e., HYSYS simulated files), the multi-objective model is formulated. All non-dominated solutions (Pareto frontier) are then generated while the convex nature of the Pareto set is also assessed. Ultimately, the WPM is used to scalarize the problem where the single optimal point of the composite function is found by using genetic algorithm.

GA is one of the most appropriate techniques in evolutionary computational optimization. It is a search heuristic technique which does not require gradient information, therefore is an effective technique regardless of the nature of the objective function (Marler and Arora, 2004). Many experience have shown the ability of this technique to converge to the global optimal point rather than local optimal solutions. In the present work, the optimization procedure is performed using an in-house coded genetic algorithm linked to the process simulator. A population size of 25 with two basic operations, i.e., crossover probability of 0.8 and mutation probability of 0.2 are chosen. The results of each generation are transferred to the MS Excel in which additional operations and calculations such as data filtering and population diversity are calculated. Diversity is an important factor in the GA in terms of avoiding premature convergence (Frederick et al., 1993; Leung et al., 1997) as well as its role as a stopping criterion (Back, 1996). Two types of diversity – being calculated in the spreadsheet – are standard deviation of fitness values in a population and Chromosome-Level diversity (or Genotypes diversity). In order to avoid premature convergence in local optima, the Social Disaster Technique (SDT) is employed after each 100 generations. In this method, a catastrophic operator is used to shift the population to an acceptable level of diversity by replacing a number of selected individuals.

4. Results and discussion

4.1. Pre-optimization and sensitivity analysis($\dot{F}_{steam}$)

In the pre-optimization phase, in order to determine the most important decision variables of the optimization problem, several case studies were performed on the RMC and GIMC. It was found

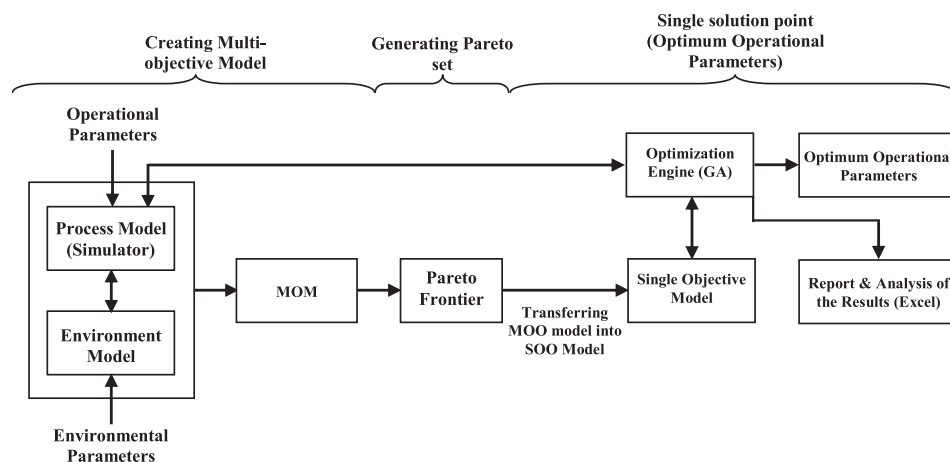


Fig. 5. Schematic procedure of the integrated methodology.

that the six following parameters had significant effect on the GIMC whereas the first four ones were the only important parameters for the RMC:

- 1 – Natural gas feed flow rate allocated for syngas production (NGFS)
- 2 – Additional injected steam mass flow rate
- 3 – Steam reformer temperature (SRT)
- 4 – Syngas recycle ratio (SRR)
- 5 – Injected carbon dioxide mass flow rate to the steam reformer reactor ($\dot{F}_{CO_2}^1$)
- 6 – Injected carbon dioxide mass flow rate to methanol synthesis reactor ($\dot{F}_{CO_2}^2$)

These decision variables directly affect the methanol production as well as the pollutants emission through the reactions taking place in the reforming and methanol synthesis sections. Fig. 6 shows these decision variables on the schematic box diagram of the proposed GIMC.

4.2. LCA and inventory analysis of RMC and GIMC

Fig. 7 shows the sensitivity of methanol production to the syngas composition (module M or S_R number) in the RMC. Four major decision variables in the RMC (i.e. NGFS, ($\dot{F}_{steam}$), SRT and SRR) have been changed so that they cover a wide range of stoichiometric ratio. Such variations lead to more than 2500 cases in which the process simulation converges. As could be observed, the maximum

amount of methanol production takes place at $M = 2.07$. This result clearly shows that the operational objective function of module $M = 2$ provided by combined steam and auto thermal reforming is applied to maximize methanol production as a desired product.

The environmental aspects of RMC could be categorized into three main sections including methanol plant, cryogenic air separation unit and its associated gas turbine power plant (GTPP) (Taghdisian et al., 2012). The environmental burdens of each section can be summarized in the atmospheric and wastewater emissions while the annual load of wastewater discharge is very low. Thus, the atmospheric pollutants emission can be considered as the main environmental aspect of the RMC (Taghdisian et al., 2012). According to the previous calculation results (Taghdisian et al., 2012), activities related to the energy generation are directly responsible for the emissions of the GHGs and air pollutants in the RMC. The GHGs and atmospheric emissions include CO_2 , CO, NO_x and VOCs (predominantly methane and methanol with the main sources being fugitive leaks). Sulfur dioxide emission from the RMC is negligible, since the natural gas feed does not contain any H_2S or other sulfur compounds. The air pollutants emission and environmental impacts of the RMC are shown in Figs. 8 and 9, respectively.

According to the inventory analysis, CO_2 is the major emitted pollutant from the RMC with an approximate annual discharge rate of 712,000 tonne/year while NO_x , CO and VOCs are the next main pollutants with the emission rates of 1904, 254 and 63 tonne/year, respectively (Fig. 6). Large amount of CO_2 emission as a greenhouse gas from the RMC introduces the global warming as its most

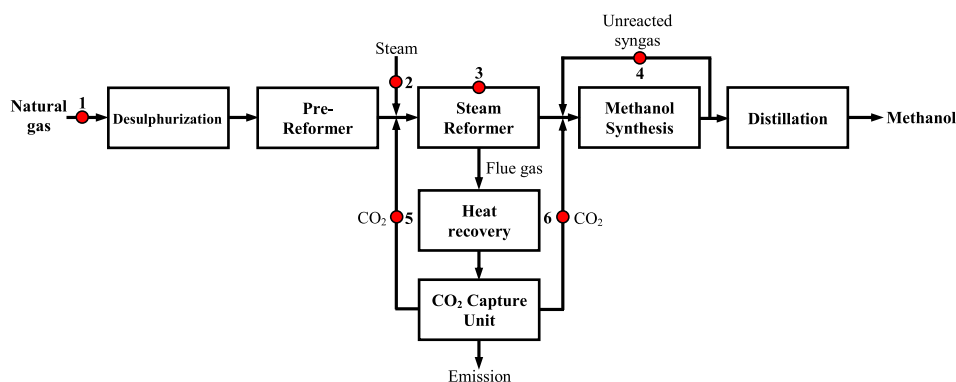


Fig. 6. Six important decision variables of the GIMC on a schematic box.

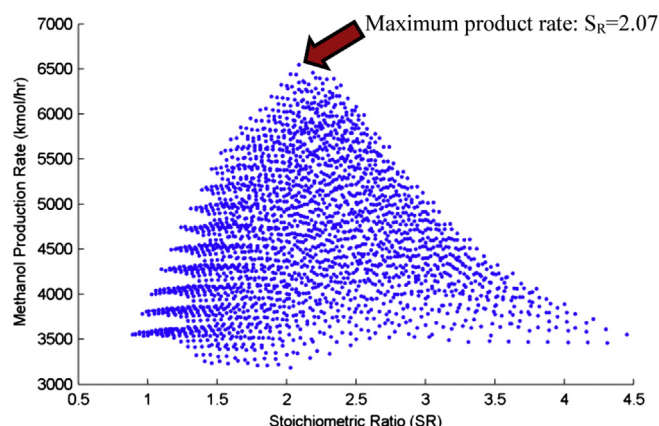


Fig. 7. Methanol production vs. module M in the reference methanol plant.

important environmental impact with an approximate amount of 760,000 tonne CO_2 equivalent/year.

As described in Section 2.2.2, in order to mitigate the CO_2 emission in the proposed GIMC, CO_2 -reforming is used where the injected CO_2 is supplied from the reformer flue gas. Since wastewater emission from CO_2 capture unit is negligible (Anna Krre et al., 2010; Peht and Henkel, 2009), the overall environmental burdens from the GIMC would be the summation of atmospheric emissions from the methanol plant and the CO_2 capture unit. The energy consumption in the CO_2 capture process consists of the thermal energy needed to regenerate solvent, the pumping electricity and the flue gas blower as well as the energy needed to compress the recovered CO_2 to the required pressure (Metz et al., 2005). Table 3 shows the important operating data of the carbon capture unit. Preliminary analysis of the GIMC and simulation results shows that, similar to the RMC, the main environmental impact of the proposed GIMC is GHGs emissions. The main emitted pollutant from the GIMC is still carbon dioxide whereas emission of the other atmospheric pollutants such as NO_x and CO is much less than that of CO_2 .

Based on the obtained results and preliminary inventory analysis of the GIMC, the multi-objective model in the proposed GIMC can be formulated as follows:

$$\begin{aligned} \text{Max } f_1(\vec{x}) &: \text{maximize Methanol production} \\ \text{Min } f_2(\vec{x}) &: \text{minimize } \text{CO}_2 \text{ Emission} \end{aligned} \quad (5)$$

The first objective function in MOM refers to classical objectives in process design in which it is tried to maximize the final product

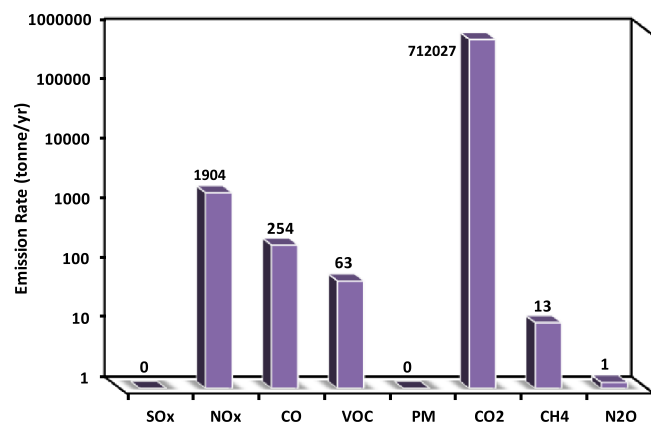


Fig. 8. Air pollutants emission from the reference methanol case.

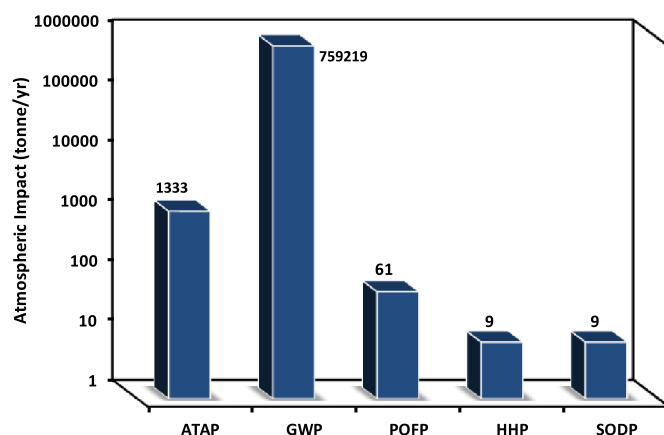


Fig. 9. Atmospheric impacts assessment for the reference methanol case.

to achieve more profit, while the second objective function refers to an environmental objective in the process design stage. It should be noted that it is possible to consider other environmental objectives in the MOM. For example, according to Fig. 8, a three objective model can be potentially developed based on the global warming and atmospheric acidification potentials as the two most important environmental impacts of the methanol process (Eq. (6)).

$$\begin{aligned} \text{Max } f_1(\vec{x}) &: \text{maximize Methanol production} \\ \text{Min } f_2(\vec{x}) &: \text{minimize GWP} \\ \text{Min } f_3(\vec{x}) &: \text{minimize AtAP} \end{aligned} \quad (6)$$

The number of objective functions is not only an important factor for the computation time, but it is also important for decision makers to choose the best compromised solution in posterior preferences methods. This is a serious problem, especially when applying evolutionary techniques to optimize problems involving a process simulator. In these cases, since the computation time needed for the simulator to converge is larger than that for the computations performed by the GA, it is suggested to choose as few as possible objective functions to develop an efficient search algorithm (Jang et al., 2005). Furthermore, in the case of using a few objective functions, namely two or three objectives, it is possible to visualize Pareto frontier in two or three-dimensional spaces (2D/3D space). Visualization of the Pareto optimal front is straightforward for decision makers as the comparison between different solutions according to their location on the Pareto frontier is easier than by other means (Madetoja et al., 2008). It should be noted that in Eq.

Table 3
Specifications of CO_2 capture unit (Metz et al., 2005).

Energy consumption in CO_2 capture unit for solvent regeneration ^a	2.7–3.3 GJ/tonne CO_2
Energy consumption in CO_2 capture unit for solvent pumping	0.2–0.3 GJ/tonne CO_2
Energy consumption in CO_2 capture unit for CO_2 compression	0.4 GJ/tonne CO_2
Total energy consumption in CO_2 capture unit	3.3–4 GJ/tonne CO_2
Chemicals needed to reclaim the amine from heat stable salt	0.03–0.13 kg NaOH/tonne CO_2
Activated carbon to remove decomposed products from solvent	0.03–0.06 kg act. carbon/tonne CO_2
Air pollutants emission from CO_2 capture unit	CO_2 , NO_x , CO
Water pollutants emission rate	Negligible

^a Values for regeneration reboiler in the stripper column depend on the solvent type.

(5), the environmental objective could be substituted by minimization of GWP instead of minimizing the CO₂ emission. In this study, CO₂ is the main greenhouse gas contributed to the global warming impact. Therefore, using CO₂ instead of GWP leads to generating an important environmental indicator in the stage of transforming MOM to SOM and that is the reason to choose Eq. (5) as our multi-objective model.

Fig. 10 indicates feasible criterion space and Pareto line for the multi-objective model based on Eq. 5. The feasible criterion space contains 44,230 data over a wide range of decision variables in which CO₂ emission (as an environmental objective function) versus methanol production (as a classical objective function) are depicted. It can be observed that the Pareto frontier is a convex line and the value of the objective functions are positive all over the feasible decision space (X) which means that WPM can be used without difficulty. Point (A) represents the coordinates relevant to the reference methanol case.

The multi-objective model based on Eq. (6), with two environmental objectives, leads to a 3D Pareto surface as shown in Fig. 11. Although it is possible for decision makers to compare different compromised solutions on the Pareto surface, decision making in 3D space is more complicated and time consuming without considerable achievement in the environmental improvement.

In order to obtain single optimal solution, the MOO model is transformed to a SOO problem by WPM (Eq. (7)) and the single optimum point is found by using GA:

$$\begin{aligned} \text{minimize } f_{eq}(\vec{x}) &= \frac{\dot{m}_{CO_2}}{\dot{m}_{MeOH}} = f(NGFS, \dot{F}_{steam}, SRT, SRR, \dot{F}_{CO_2}^1, \dot{F}_{CO_2}^2) \\ \text{s.t.} \\ 0 \leq \dot{F}_{CO_2}^1 &\leq 110 \text{ tonne/h} \\ 0 \leq \dot{F}_{CO_2}^2 &\leq 66 \text{ tonne/h} \\ 850^\circ\text{C} \leq SRT &\leq 1000^\circ\text{C} \\ 1.2 \leq \frac{\text{Steam}}{\text{Carbon}} &\leq 2.5 \\ SRR \leq 0.95 \end{aligned} \quad (7)$$

The composite function (f_{eq} or CO₂-efficiency indicator) which should be minimized represents the amount of the emitted carbon dioxide (mass basis) per unit mass of methanol production. This

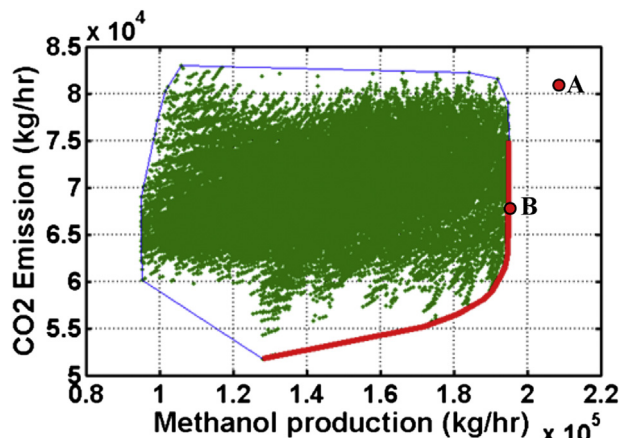


Fig. 10. Feasible criterion space and convex Pareto line of multi-objective model based on equation (5). Points (A) and (B) represent the optimal condition of the RMC and GIMC, respectively.

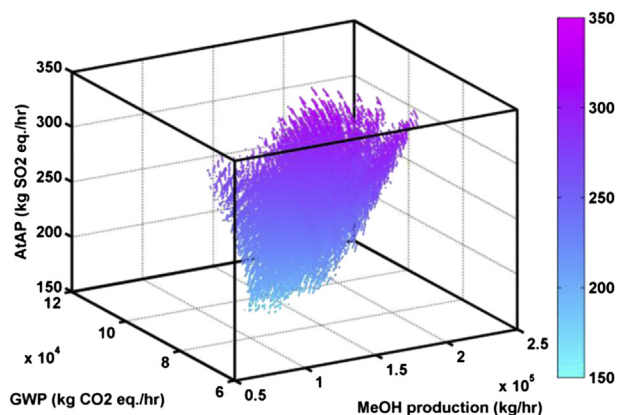


Fig. 11. Pareto surface for three objective model based on Eq. (6).

index is based on Eco-efficiency concept in which CO₂ emission in the numerator refers to “environmental influence” while the rate of methanol production in the denominator is a representative for “product and services”. In order to properly compare the GIMC and RMC, the objective function is subjected to several constraints. The first two constraints indicate that the amount of injected carbon dioxide is needed to be within the given ranges so that the actual volume flow passing through equipments remains in the allowable range of the RMC. Based on several performed case studies and preliminary analyses of the GIMC, the steam reformer outlet and the methanol synthesis reactors outlet have the maximum volumetric flow among all process flows in the GIMC. Therefore, if these actual volumetric flows are kept within the allowable range of the RMC, actual volumetric flows through all over the GIMC will remain in the allowable range of the RMC. Steam reformer temperature should also be kept within the given range to maintain acceptable conversion of methane to syngas as well as to avoid coke formation. Steam to carbon ratio is an operational parameter dictated by catalyst manufacturer to avoid coke deposition on reforming catalysts (Demidov et al., 2011; Huang and T-Raissi, 2007; Lange, 2001; Li et al., 2008). The last constraint – syngas recycle ratio – is required to be less than 95% as the computation time for converging the simulator would be very long out of this range.

Fig. 12 shows fitness values of Eq. (7) in each generation of GA. The best fitness value converges to the value of 345 kg CO₂/tonne MeOH which means that CO₂ emission of the GIMC is improved about 45 kg CO₂/tonne MeOH (or 11.5%) with respect to the RMC. Table 4 shows the values of the decision variables at the optimum point. In Fig. 10, point (B) represents the CO₂ emission versus the

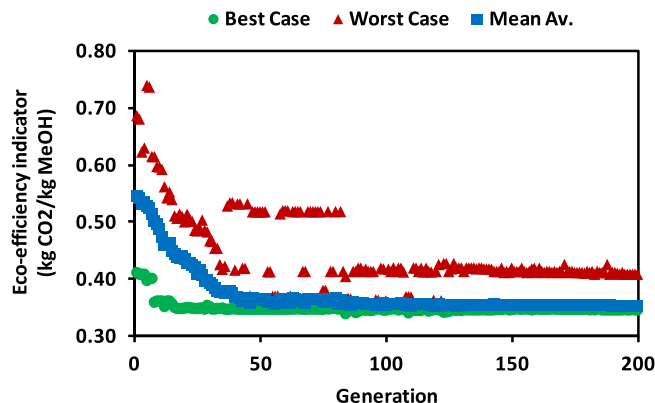


Fig. 12. The best, average and worst fitness values of Eq. (7) in each generation.

Table 4

Values of the decision variables and objective function at the optimum point of GIMC.

NGFS	95.4 tonne/h
($\dot{F}_{\text{steam}}$)	22.8 tonne/h
SRT	983 °C
SRR	93.8%
($\dot{F}_{\text{CO}_2}$)	33 tonne/h
($\dot{F}_{\text{CO}_2}$)	53.7 tonne/h
Obj. Fcn : $f_{\text{eq}}(\vec{x}) = \dot{m}_{\text{CO}_2}/\dot{m}_{\text{MeOH}}$	345 kgCO ₂ /tonne MeOH

methanol production at the optimal condition of the GIMC. It is observed that the optimal condition obtained from WPM lies on the Pareto frontier. Comparing points (A) and (B) implies that the optimal condition of the RMC which is based on a traditional objective (i.e. maximizing methanol production) is a degenerate solution from the view point of environmental benign design, whereas by involving the environmental objective in the design stage, a compromised optimum could be achieved. Coordination of the optimal point of the GIMC clearly shows the trade-off between the environmental and classical objectives in the multi-objective approach where the CO₂ emission as an environmental objective improves with a decrease in the methanol production.

In order to assess premature convergence and trapping in a local optimal point, population diversity was calculated. Fig. 13 shows GA diversity based on the standard deviation of the fitness values in the population. The result displays good population diversity through converging procedure. This acceptable diversity can be clearly seen in Fig. 14 where the best fitness value remains the same even after implementing social disaster technique (SDT) in every 100 generations. The figure indicates that searching process has not been trapped in local optima.

Fig. 15 shows the methanol production as a function of module M for 200 generations during the GA search process. As can be seen, by introducing the environmental objective in the design stage, the maximum methanol production of 196 tonne/h occurs at module $M = 1.92$ compared to 208 tonne/h at $M = 2.07$ in case of the RMC. Indeed, this result was expected, because the lower value of M (1.92 in MOO) implies that, by involving the environmental objective in the optimization procedure of the proposed GIMC, CO₂ emission with respect to the RMC could be reduced about 13.5 tonne/h at the expense of 12 tonne/h decrease in the methanol production. It corresponds to the reduction of 16.6% in the CO₂ emission along with 5.7% decrease in methanol production both with respect to the

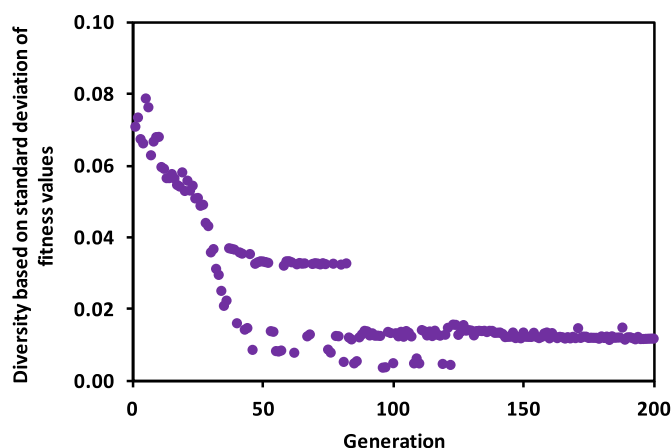


Fig. 13. GA diversity based on the standard deviation of fitness values in the population.

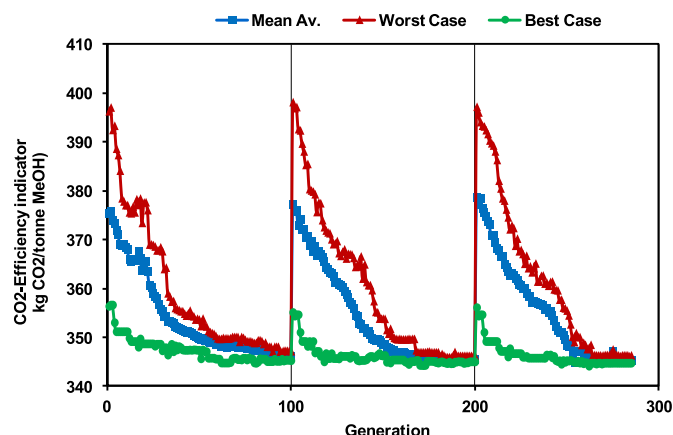


Fig. 14. Implementation of social disaster technique after every 100 generations.

RMC. The main operational data of the RMC and GIMC at their optimum points are compared in Table 5.

5. Conclusion

Two different approaches for the methanol production were compared from the view point of eco-design. The results showed that involving the environmental objectives along with the classical objectives in the process design stage yields a different optimal solution. In fact, without considering environmental regulations as a separate objective in the process optimization, the optimal point would be the local optimal solution whereas in the multi-objective optimization approach, the optimal point is a global optimal solution. It was found that using the multi-objective approach in the proposed GIMC, leads to a lower CO₂ emission (as the main environmental objective) with respect to the RMC at the expense of a small decrease in the methanol production. Although CO₂-efficiency, as an environmental indicator, could be used to compare the environmental performance of different industries, it is hard to be interpreted by non-environmentally trained managers. Therefore, in our future work, the weighted sum method is employed to solve the multi-objective optimization problem with the environmental objective functions considered as economic criteria, which is a more familiar term for traditional decision makers. This method enables decision makers to trade-off between environmental impacts and economic cost in the same monetary unit which help them to select a more environmentally and economically desirable design.

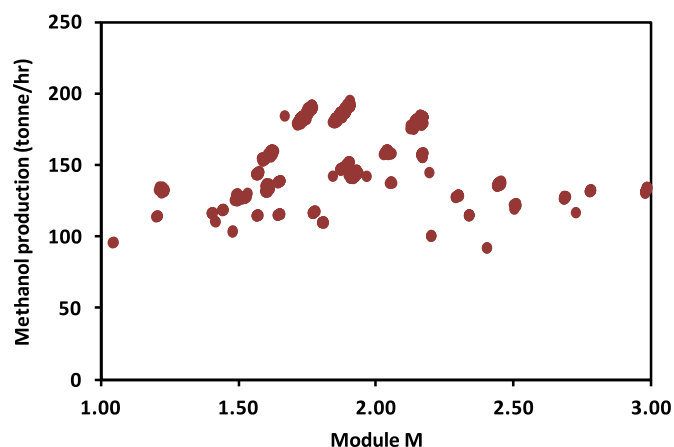


Fig. 15. Methanol production vs. module M during GA search for 200 generations.

Table 5

Comparison of the main operational data of the RMC and GIMC at their optimum points.

Operational/process data	RMC (single objective approach)	GIMC (multi-objective approach)
Syngas temperature (°C)	975	983
Injected CO ₂ (tonne/h)	—	86.8
NGFS (tonne/h)	127.9	95.4
Additional injected steam (tonne/h)	—	22.8
Syngas recycle ratio (%)	95	93.8
Module M (or S _R)	2.07	1.92
Steam/carbon at steam reformer inlet (mol %)	1.71	1.71
Syngas flow rate at reformer outlet (tonne/h)	259	208
Methanol production (tonne/h)	208	196
CO ₂ -efficiency (kg CO ₂ /tonne MeOH)	390	345

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